



MCQs – Software Construction

Lecture 6: Specifications

(Dr. Ehab Essa Exact Exam Style)

1. Why are software specifications important in software construction?

- A. Improve UI
- B. Define behavior
- C. Reduce memory
- D. Speed execution

Answer: B

2. A specification mainly describes:

- A. Implementation
- B. Algorithm steps
- C. Expected behavior
- D. Source code

Answer: C

3. Specifications act as a contract between:

- A. Compiler and JVM
- B. Tester and debugger
- C. Client and implementer
- D. OS and hardware

Answer: C

4. Which bugs usually arise without clear specifications?

- A. Syntax bugs
- B. Interface misunderstandings
- C. Memory leaks
- D. Performance issues

Answer: B

5. Specifications help the client because they:

- A. Improve speed
- B. Avoid reading code
- C. Reduce bugs
- D. Increase coupling

Answer: B

6. Specifications help the implementer because they:

- A. Fix bugs
- B. Remove testing
- C. Allow implementation change
- D. Enforce syntax

Answer: C

7. The concept that allows independent change of client and implementation is:

- A. Coupling
- B. Firewall
- C. Inheritance
- D. Polymorphism

Answer: B

8. Behavioral equivalence means:

- A. Same code
- B. Same output always
- C. Interchangeable for client
- D. Same performance

Answer: C

9. Two methods are behaviorally equivalent if:

- A. They use same algorithm
- B. Client cannot distinguish them
- C. They return same value always
- D. They have same name

Answer: B

10. A method specification consists of:

- A. Header and body
- B. Input and output
- C. Precondition and postcondition
- D. Name and parameters

Answer: C

11. The precondition is an obligation on the:

- A. Compiler
- B. Tester
- C. Client
- D. Implementer

Answer: C

12. The postcondition is an obligation on the:

- A. Client
- B. Implementer

- C. JVM
- D. OS

Answer: B

13. If a precondition does NOT hold, the method:

- A. Must return null
- B. Must throw exception
- C. Is not bound by postcondition
- D. Must terminate

Answer: C

14. Design by Contract includes:

- A. Preconditions only
- B. Postconditions only
- C. Preconditions, postconditions, invariants
- D. Comments only

Answer: C

15. In Java, specifications are usually written using:

- A. Annotations
- B. Assertions
- C. Documentation comments
- D. Keywords

Answer: C

16. Which Java tag is used to describe parameters in specs?

- A. @return
- B. @throws
- C. @param
- D. @spec

Answer: C

17. A specification should NOT mention:

- A. Parameters
- B. Return value
- C. Local variables
- D. Exceptions

Answer: C

18. Test cases must follow:

- A. Implementation
- B. Specification
- C. Debugger
- D. Compiler

Answer: B

19. A test case that violates the precondition is:

- A. Valid
- B. Required
- C. Incorrect
- D. Optimized

Answer: C

20. Unit testing focuses on:

- A. Whole system
- B. UI behavior
- C. Single specification
- D. Performance

Answer: C

21. Integration testing mainly checks:

- A. Syntax
- B. Individual methods
- C. Compatibility of specs
- D. Memory usage

Answer: C

22. A deterministic specification:

- A. Allows many outputs
- B. Has one valid output
- C. Is unclear
- D. Is operational

Answer: B

23. A specification allowing multiple valid outputs is:

- A. Deterministic
- B. Declarative
- C. Underdetermined
- D. Strong

Answer: C

24. Declarative specifications describe:

- A. Steps of execution
- B. Algorithm logic
- C. Properties of result
- D. Loop structure

Answer: C

25. Operational specifications mainly describe:

- A. Final result
- B. Output relation
- C. Execution steps
- D. Abstraction

Answer: C

26. A stronger specification has:

- A. Stronger precondition
- B. Weaker postcondition
- C. Weaker precondition
- D. Fewer guarantees

Answer: C

27. Replacing an old specification with a new one is safe if:

- A. Precondition is stronger
- B. Postcondition is weaker
- C. Precondition is weaker
- D. Behavior is hidden

Answer: C

28. A good specification should be:

- A. Long
- B. Vague
- C. Clear and coherent
- D. Algorithmic

Answer: C

29. Using abstract types in specifications increases:

- A. Coupling
- B. Restrictions
- C. Flexibility
- D. Complexity

Answer: C

30. Making helper methods private mainly improves:

- A. Speed
- B. Interface coherence
- C. Memory
- D. Compilation

Answer: B